ARNAV VERMA

Dehradun, India | arnav04verma@gmail.com | Portfolio | Upwork | Linkedin | GitHub | Medium

PROFESSIONAL SUMMARY

Embedded Systems Engineer specialising in **power electronics, PCB design, and embedded firmware**, with additional experience in **autonomous UAV systems** (**PX4, ArduPilot, ROS 2**). Delivered production-ready **KiCad** designs, including isolated SMPS and USB-isolated interface boards, for international freelance clients, from schematic and component selection through Gerbers, BOM, and manufacturing handoff. Additional background in **RF/wireless systems** (**LoRa FHSS, SDR**) and embedded **C/C++ firmware** across **STM32/ESP32** platforms.

Education

Thapar Institute of Engineering and Technology, BE in Electronics and Computer — Aug 2023 – Aug 2027

- CGPA: **9.14/10.0** | Digital System Design, Analog Electronics, Computer Architecture, Embedded Systems, DSP

Work Experience

Autonomous Drone Engineer (Contract), NinjaTech — August 2026 – Present

- Developing autonomous UAV solutions.
- Supporting embedded systems development and integration.

Embedded Systems and UAV Engineer, UPWORK, Freelance — April 2026 – Present

- Designed PCBs, embedded firmware, and Gazebo-based UAV simulations for international clients.
- Assembled and integrated UAV electronics and RF communication modules.

UAV Systems Intern, EME corps, Indian Army — March 2026

- Completed a 3-week internship with the EME Corps, contributing to a Hailing Swarm Drone System
- Awarded a **Certificate of Internship** by EME for successful project delivery

Head of Avionics, Team Sammpaati, TIET — August 2025 – Present

- Led a 10-member team for the NIDAR Disaster-Management Drone Competition (DFI), developing and integrating an autonomous UAV platform.
- Conducted UAV workshops and trained **40+ juniors** in embedded systems and communication protocols

Projects

Isolated USB Radio-to-PC Interface — *Project Link*

- Designed a fully-isolated USB-to-radio interface board for a commercial telecom client: **ADuM3160** USB isolation, **CM119B** audio codec, **FT231X** UART, and dual-OS PTT/COR control (Linux HID GPIO + Windows FTDI modem lines) for drop-in Asterisk `chan_usbradio` compatibility.
- Delivered a manufacturing-ready **KiCad** package (schematic, PCB, Gerbers, BOM/CPL, panel drilling drawings) for JLCPCB fab, resolving isolation-barrier creepage/clearance and **thermal/signal-integrity** considerations; verified **alternate/second-source components** (CM108 family, genuine-vs-counterfeit FTDI) for supply resilience.

Universal Isolated SMPS PCB — *Project Link*

- Designed a **30W**, universal-input (**85–265 VAC**) offline flyback SMPS in **KiCad 9** around a **TOP244Y** controller, **4kV** isolation, EMI/surge protection, and isolated feedback; produced manufacturing-ready Gerbers with **12V** output.

Fully Autonomous Heavy-Lift UAV for Disaster Response — *Flight Video*

- Designed, manufactured, and flight-tested a custom coaxial UAV with a **1000mm wheelbase**, hybrid 3D-printed and carbon-fiber tube airframe, using **18” rotors** to achieve **22 min endurance** and **5 kg payload capacity**
- Integrated **Pixhawk** + **Raspberry Pi** companion computer; built KML-driven autonomous missions (area scan, survivor detection, payload delivery) and an onboard **OpenCV + ML** inference pipeline for real-time flood-survivor detection

Custom FHSS Radio Link for UAV Swarms — *Project Link*

- Built a custom **ESP32 + SX1278 LoRa FHSS** communication system for reliable **UAV swarm** telemetry.
- Implemented **frequency hopping**, synchronization, and embedded communication protocols; verified RF performance with **HackRF One.**

Skills

Languages: C, C++, Python
Embedded Systems:STM32, ESP32, Arduino, UART, SPI, I2C, PWM, CAN, RS485
Lab Equipment: Oscilloscope (DSO/MSO), Function Generator, DC Power Supplies, Multimeter
Companion Computers: Raspberry Pi 4/5, ROS2, Linux, Shell scripting, Jetson Nano, JetBot
Drone Autonomy and Control: PX4, Ardupilot, MAVLink/MAVSDK, DroneKit, SITL, Gazebo, Python
Communication and Networking: Wi-Fi, Bluetooth, LoRa, Adalm Pluto SDR, HackRF One, GNU Radio, GQRX
Tools and Platforms: KiCad, MAVProxy, QGroundControl, Mission Planner, PlatformIO